
2026 International Conference on Intelligent Systems in Engineering, Secured Systems and Cybersecurity : Submission (1229) has been edited.

1 message

Microsoft CMT <noreply@msr-cmt.org>
To: msswativashisht@gmail.com

Fri, Apr 17, 2026 at 9:35 AM

Hello,

The following submission has been edited.

Track Name: ICISESSC2026

Paper ID: 1229

Paper Title: Deep learning system based on YOLO for finding insects in real time in precision agriculture

Abstract:

Managing pests on farms is an important part of Food security and modern ways to protect crops. This study introduces a YOLO-based deep learning framework for real-time agricultural pest detection utilizing the YOLOv8 architecture Roboflow exported an annotated pest dataset that had five pest categories: weevil, caterpillar, grasshopper, slug, and snail. The suggested system's mean Average Precision (mAP@0.5) for all pest classes is 0.952. The AP scores for each class range from 0.849 for caterpillars to 0.995 for snails and grasshoppers. To make sure that the framework works well in a variety of field situations, It uses data augmentation methods, decoupled anchor-free detection heads, and multi-scale feature pyramid networks. Tests on Google Colab's GPU infrastructure show that the system works much better than baseline models like YOLOv5 and Faster R-CNN, which can process more than 60 frames per second in real time with good accuracy and speed. Some uses are automated systems for monitoring crops, drones for field surveillance, and platforms for precision agriculture, and smart farms can get help and advice. Subsequent research will focus on expanding the framework to include additional crop and pest species, adding more advanced versions of YOLO and making it work on edge devices with limited resources.

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Primary Subject Area: AI, Data Science & IoT Applications

Secondary Subject Areas: Not Entered

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Submission Questions Response: Not Entered

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